

Computational Optical and Bioimage Analysis for Quantitative Measurement

A classical, auditable segmentation and measurement pipeline
on the 2018 Data Science Bowl nuclei dataset (BBBC038)

Abstract

This report describes a reproducible, deliberately classical workflow for measuring objects in light-microscopy images: load, preprocess, threshold, split touching nuclei by a distance-transform watershed, extract per-object geometry, and then quantify how far those measurements can be trusted as image quality degrades. On a held-out 470-image portion of the BBBC038 `stage1_train` set the pipeline reaches a pooled pixel Dice of 0.872 but a per-object F1 at IoU 0.5 of only 0.784, with 1,814 merge events — and the macro-average over acquisition modalities falls further, to Dice 0.781 and F1@0.5 0.641. That gap between a pixel score and an object score is the substantive result: pixel overlap is nearly blind to the failure mode classical segmentation actually has, which is getting the object *count* wrong. A controlled degradation sweep over noise, blur and contrast then shows which measurements survive a worse image, including one exactly flat null result that follows from Otsu’s affine equivariance rather than from a failed experiment.

1 Dataset

The data is the 2018 Data Science Bowl nuclei collection, distributed as BBBC038 by the Broad Bioimage Benchmark Collection. Only the `stage1_train` archive is used, because it is the only one with public instance-level ground truth: each nucleus is a separate binary PNG, which is what makes object counting and per-object precision and recall possible at all. Merging those per-nucleus masks into one `int32` label image per field of view gives 29,461 annotated nuclei across 670 images, at 9 distinct image sizes, with between 1 and 375 nuclei in a single field.

The collection is heterogeneous by design, and two properties of it drive every later choice. First, polarity is mixed: 124 of the 670 images are dark nuclei on a light background (histology and brightfield), and the rest are bright nuclei on a dark field. A naive Otsu pipeline silently segments the *background* on those images while still reporting a plausible Dice, so polarity is detected once per image, recorded in `data/manifest.csv` with a hand-override column, and never re-derived. Second, 108 images carry real colour while the remainder are effectively grayscale.

Images are grouped by a two-question rule — is the image coloured, and does it have dark nuclei on a light

field — rather than by a fitted clustering, because a rule can be read in one sentence, explained, and corrected by hand in the manifest. The rule admits four groups; `stage1_train` populates three of them, with no colour-fluorescence images at all (Table 1).

A held-out split is not optional here. Four parameters (watershed h , minimum object area, CLAHE clip limit, denoise radius) were tuned, and tuning and reporting on the same images would make every number an in-sample estimate. The split assigns an image by a hash of its identifier, so it depends on neither file ordering nor a random number generator’s version: an image lands in the same split forever.

2 Methods

Preprocessing. Each image is loaded as RGBA, its constant alpha channel dropped after asserting that it really is constant, and reduced to one grayscale channel — by taking a single channel when all three agree, and by `rgb2gray` otherwise. Images are never resized: resizing changes object scale in pixels, which would make a minimum-area threshold mean different things on different images. The pipeline then applies, in order, polarity normalization from the manifest, white-tophat background subtraction with a structuring element comfortably larger than the largest nucleus, a small median filter (which suppresses speckle without moving edges as a Gaussian would), and CLAHE with a deliberately low clip limit, because a high one amplifies histology background texture into false objects.

Segmentation. Three methods are compared on identical preprocessed input: a global Otsu threshold, a local adaptive threshold, and a marker-controlled watershed. The watershed is the recipe that matters: remove small objects and holes, compute the Euclidean distance transform, smooth it by one pixel, take `h_maxima` of the smoothed transform as markers, label those markers with connected components, and flood `watershed(-dist, markers, mask=binary)`. The one-pixel smoothing is not cosmetic — the distance transform of a binary blob is full of exactly-tied plateaus, and `h_maxima` on the raw transform returns one marker per tie group instead of one per nucleus. The depth h is expressed in distance-transform pixels, which makes it directly interpretable: two nuclei are separated when the neck between them is at least h pixels narrower in radius than the smaller of the two.

Table 1: Composition of the dataset after grouping and splitting. Tuning used the dev split only; every result reported below is measured on test.

Modality group	Images	Dev	Test	Nuclei	Inverted
Grayscale fluorescence	546	171	375	23,541	0
Colour histology	103	26	77	4,486	103
Grayscale brightfield	21	3	18	1,434	21
Total	670	200	470	29,461	124

Border objects are deliberately *not* cleared, because the ground truth includes partial nuclei at the frame edge, and one connectivity (8-connected) is applied identically to ground truth and prediction.

Metrics. The intersection-over-union between every ground-truth object and every predicted object is built in one vectorized pass over a contingency table, which at this scale is the difference between seconds and hours. Matching is greedy, which here is exact rather than approximate: at an IoU threshold of 0.5 or above an object cannot exceed the threshold against two different objects, so the matching is unique. Reported quantities are pixel Dice and IoU, per-object F1 at IoU 0.5 and 0.7 computed the same way the Data Science Bowl paper computes them, the official DSB score (the mean of $TP/(TP + FP + FN)$ over IoU 0.50 to 0.95 in steps of 0.05), split and merge counters, and both signed and absolute count error. Every table is broken down per modality group with a macro average, because the groups are badly imbalanced and a dataset-wide mean would hide total failure on the smallest.

Verification without a test suite. The two plausible failures here — polarity inversion and watershed splitting or merging — are both silent: they produce a believable Dice while the result is wrong. Instead of a test framework, the pipeline begins by scoring four synthetic cases with known answers and printing them (Table 2). The second row is the whole argument of this project in one line.

3 Results

3.1 Pixel agreement and object agreement disagree

Table 3 is the headline result. Pooled over all 470 test images the watershed reaches Dice 0.872 — pixel-level agreement that would read as a solved problem — while per-object F1@0.5 is 0.784, F1@0.7 is 0.663, and the official DSB score is 0.475. The pipeline recorded 1,814 merges against only 290 splits, and predicted 18,704 nuclei where the ground truth has 20,729. In other words the segmenter finds the nuclear material accurately and then fails to divide it into the right number of objects, mostly by fusing neighbours. Dice is almost blind to this, which is exactly what the self-check demonstrates on a synthetic image.

The gap between the macro and pooled rows is the second thing worth reading. Macro Dice is 0.781 against pooled 0.872, and macro F1@0.5 is 0.641 against pooled 0.784. Grayscale fluorescence — small, round, well-separated nuclei on a dark field — scores Dice 0.891 and F1@0.5 0.828. Colour histology holds its pixel score at Dice 0.841 but drops to F1@0.5 0.643: dense tissue means touching nuclei, and touching nuclei are what a distance-transform watershed gets wrong. Grayscale brightfield is the weakest group on both axes, at Dice 0.610 and F1@0.5 0.452, with a mean absolute count error of 18.833 nuclei per image. It is also the smallest group (21 images), so its numbers are the least stable, but the failure is structural rather than statistical: brightfield nuclei have low contrast against cytoplasm and soft boundaries, so a global threshold has no clean level to find and the distance transform inside a poorly-delimited blob offers the watershed no reliable peak to use as a marker.

Comparing the three segmenters on identical input shows where the watershed earns its place. Otsu and watershed share a threshold and therefore essentially the same macro Dice (0.780 against 0.781), but splitting fused blobs lifts macro F1@0.5 from 0.568 to 0.641. The adaptive threshold is worse on both counts here (macro Dice 0.611, F1@0.5 0.403), because a local threshold finds structure inside a dark background as readily as it finds nuclei.

3.2 Against the published baselines — and why this is not a win

Because F1@0.5 and the official DSB score are computed with the DSB paper’s own definitions, our numbers can be placed beside its published leaderboard (Suppl. Table S3 of the accompanying metadata). Table 4 does this, and the “Scored on” column is part of the result rather than a footnote.

Read naively, our pooled F1@0.5 of 0.784 sits above the published U-Net entry at 0.734. That comparison is invalid and we do not claim it. The published entries are scored on the competition’s held-out test sets, assembled to include acquisition types and difficulties absent from stage1_train; our split is a hash-based partition of stage1_train itself, which is both easier and dominated by small fluorescent nuclei — roughly 546 of 670 images, the easiest category in the collection. The honest comparison is the official DSB score, which penalizes at every IoU threshold from 0.5 to 0.95 and is far less forgiving of approximately correct boundaries. There we reach 0.475 pooled and 0.331 macro, against

Table 2: Known-answer self-check, printed at the start of every run. A segmentation that cuts every nucleus in half is pixel-perfect and object-wise worthless, and only the object metrics can see it.

Synthetic case	Dice	F1@0.5	Splits	Merges
identical to ground truth	1.00	1.00	0	0
every nucleus split in two	1.00	0.00	2	0
both nuclei merged into one	0.67	0.00	0	1
no overlap at all	0.00	0.00	0	0

Table 3: Watershed on the held-out test split, per modality group. The macro average weights each acquisition type equally; the pooled row is dominated by grayscale fluorescence, which is about 546 of the 670 images.

Group	Images	Dice	F1@0.5	F1@0.7	DSB	Splits	Merges
Grayscale fluorescence	375	0.891	0.828	0.718	0.529	163	1,198
Colour histology	77	0.841	0.643	0.470	0.278	121	456
Grayscale brightfield	18	0.610	0.452	0.359	0.185	6	160
Macro average	470	0.781	0.641	0.515	0.331	290	1,814
All images (pooled)	470	0.872	0.784	0.663	0.475	290	1,814

0.638 for U-Net, 0.589 and 0.724 for CellProfiler configured by a novice and by an expert, and 0.820 for the winning entry — below every published entry, by a wide margin, on the metric the competition itself used. That is the expected position for a transparent classical baseline, and what this project set out to measure rather than to beat.

3.3 Per-object features

Applying `regionprops` to the watershed labels, with the original grayscale as the intensity image, yields 26,112 measured nuclei, plus two derived scale-invariant columns: circularity $4\pi A/P^2$ and aspect ratio. Orientation is excluded — for an approximately round object it is an arbitrary number.

A PCA over all morphology features gives the guaranteed and uninteresting answer: PC1 explains 0.555 of the variance and loads with the same sign on area, perimeter, both axis lengths, equivalent diameter and maximum Feret diameter. Six columns measuring the same thing produce one component meaning “size”. Over scale-invariant shape features only, PC1 at 0.659 is instead a roundness axis, with circularity, solidity and extent opposing eccentricity and aspect. KMeans on the leading shape components selects $k = 2$ by silhouette (0.479), separating round compact nuclei from elongated irregular ones. Because that split partly tracks acquisition type, the clustering was repeated inside the largest modality group alone, where the same two-cluster structure survives — so the shape variation is not purely an artifact of which microscope took the picture.

3.4 Measurement reliability under degradation

The final stage degrades a stratified sample of images along four axes — additive Gaussian noise, Poisson shot noise, Gaussian blur, and contrast compression — and re-runs the entire pipeline on each degraded

image. Degradations are applied to the *raw* image before preprocessing, because the pipeline exists to cope with bad images and has to be allowed to react to them; degrading after normalization would measure nothing physical. Each cell draws its noise from a seed derived from the image, axis and level, so cells are independent and adding a level does not re-roll the others.

Noise. Grayscale fluorescence tolerates additive noise up to about $\sigma = 0.05$ and then collapses, from 0.922 at $\sigma = 0$ to 0.467 at $\sigma = 0.2$. Brightfield fails far earlier and far harder, reaching 0.082 — its nuclei were barely above the background to begin with, so a noise floor at a fifth of the intensity range erases them entirely. Colour histology is the most robust of the three (0.760 to 0.674), because stained nuclei carry a large intensity separation from the surrounding tissue. Poisson noise produces the same ordering: fluorescence is almost unaffected down to a peak of 10 photons (0.898) while brightfield falls to 0.015.

Blur. All groups degrade under blur, and a single σ in pixels is not a fair insult across groups: nucleus radii in this collection span roughly 5 to 30 pixels, so $\sigma = 4$ is mild for a large histology nucleus and annihilating for a small fluorescent one. The repository therefore also plots blur against σ divided by each image’s own median nucleus radius.

Contrast: a flat null result. The contrast axis produced no effect at all. Across all 6 levels, down to a compression factor of 0.05, the median F1@0.5 in every group is identical to its baseline — the largest spread within any group across the whole axis is 0.0000. This is not a broken experiment; it is a property of the method. Otsu’s threshold is equivariant under an affine intensity transform, and the pipeline works in float32, so compressing contrast about the image median has no route by which to change the segmentation: neither

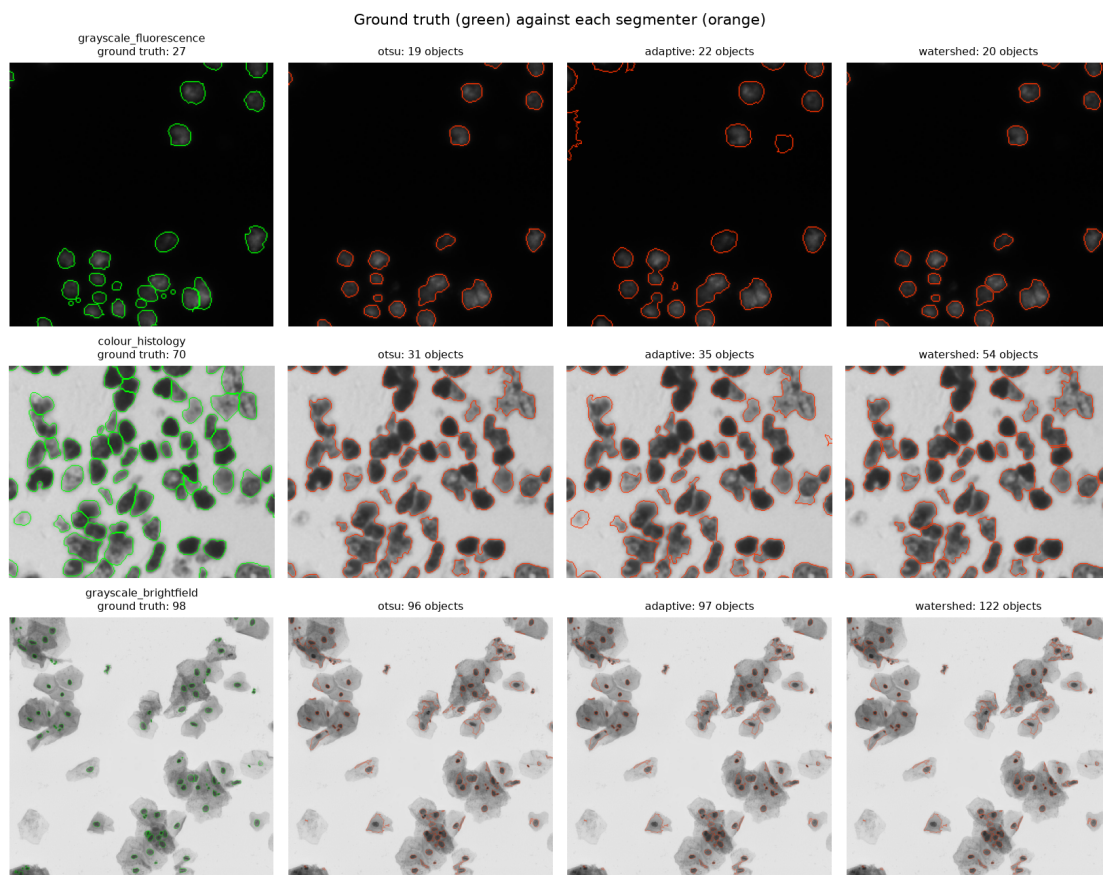


Figure 1: Ground truth and the three segmentations on one example from each modality group. Boundary overlays are drawn with the watershed line enabled for visibility; all reported metrics use the version without it, because a one-pixel boundary costs real Dice and can shrink small nuclei below the minimum-area threshold.

clipping nor quantization intervenes. Reporting a flat curve and explaining it is the correct outcome, and it also tells us what the informative experiment is.

Contrast only matters when something else does not shrink with it. Compressing the contrast and *then* adding a fixed $\sigma = 0.02$ of noise shrinks the signal while leaving the noise floor where it was, and the result (Table 5) is a steep collapse: F1@0.5 falls from 0.705 at full contrast to 0.489 at a compression factor of 0.15 and 0.053 at 0.05, against 0.715 everywhere along the noise-free row. The single-factor contrast experiment measures nothing; the interaction is the real effect, and the ratio between signal and noise floor — not contrast itself — is the physical quantity that governs it.

Separating what drifted. Degrading an image changes both the segmentation and the intensity measurements, so end-to-end drift alone cannot say which caused it. Each degraded image is therefore measured twice — once with the predicted mask (what actually happens) and once with the ground-truth mask (pure feature sensitivity, segmentation held fixed) — and the difference is the segmentation’s contribution. One analytic check validates the harness: under additive $N(0, \sigma)$ noise, per-object `intensity_mean` should be unbiased while `intensity_std` inflates as $\sqrt{s^2 + \sigma^2}$, and the measured values track that prediction closely

until the highest level, where clipping at the intensity bounds begins to bite. The consequence constrains interpretation: intensity features are not comparable across noise levels, so robustness conclusions must rest on morphology and segmentation. Repeating the stochastic axes three times at two levels confirms the curves measure differences between images rather than the luck of a random draw — the spread between draws of one image is between 0.063 and 0.142 of the spread between images.

4 Discussion

The result this project exists to report is the disagreement between two ways of scoring one segmentation. A pooled Dice of 0.872 and a pooled F1@0.5 of 0.784, with 1,814 merges, describe the same pipeline; anyone reading only the first would conclude the problem is essentially solved, while anyone counting nuclei with it would be short by 7.326 nuclei per image on average. For a measurement workflow, as opposed to a pixel-labelling benchmark, the object metrics are the ones that correspond to the question asked, and pixel overlap is the diagnostic that cannot fail loudly.

The per-group breakdown carries the second lesson. A dataset-wide average over a collection of 546 easy images and 21 hard ones describes neither; the macro average is lower than every pooled figure precisely

Table 4: Published DSB baselines at IoU 0.5 against this pipeline. **These are not like-for-like.** The published rows score the DSB stage1 and stage2 test sets; our rows score a held-out portion of stage1_train.

Method	Scored on	F1@0.5	Official DSB score
[ods.ai] topcoders	DSB test sets	0.890	0.820
Jacobkie	DSB test sets	0.870	0.811
Deep Retina	DSB test sets	0.860	0.787
CellProfiler - Expert	DSB test sets	0.819	0.724
U-Net models	DSB test sets	0.734	0.638
CellProfiler - Novice	DSB test sets	0.705	0.589
This work – watershed, pooled over test images	stage1_train held-out	0.784	0.475
This work – watershed, macro over groups	stage1_train held-out	0.641	0.331

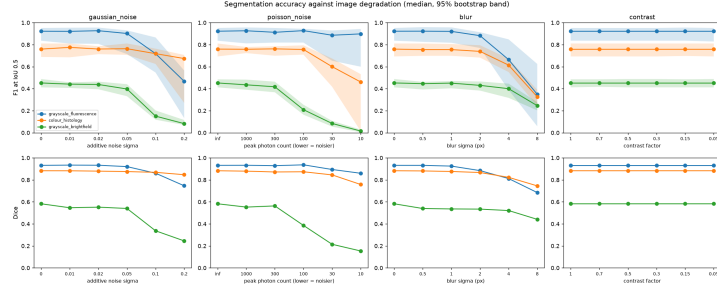


Figure 2: Median F1@0.5 against degradation level, one line per modality group, with a percentile bootstrap 95% band on the median.

Table 5: The two-factor grid: contrast compression (columns) crossed with a fixed amount of additive noise applied afterwards (rows). The top row is the flat contrast axis; the rows below it are where contrast loss actually matters.

Added noise σ	1	0.7	0.5	0.3	0.15	0.05
0	0.715	0.715	0.715	0.715	0.715	0.715
0.02	0.705	0.706	0.718	0.704	0.489	0.053
0.05	0.694	0.683	0.600	0.358	0.069	0.018

because it refuses to let the largest group speak for the rest.

The robustness stage converts “the method is sensitive to image quality” from an assertion into a measurement with a specific shape: noise is tolerated well by fluorescence and badly by brightfield; blur damages everything relative to nucleus size; contrast alone does nothing at all to an affine-equivariant threshold; and contrast crossed with noise is devastating. That last pair is the actionable finding, because in a real microscope contrast loss and a fixed detector noise floor arrive together.

5 Limitations and future work

The split is a hash-based partition of `stage1_train` rather than the competition’s own test sets, so the absolute numbers are optimistic relative to any published DSB result (Section 3.2), and grayscale brightfield has so few images that its ordering is robust while its exact values are not. Parameters were tuned by one-dimensional sweeps, not jointly, and the watershed uses a single h across a dataset whose nucleus radii span a factor of six — visible in the dev sweep as a shallow, flat optimum with no depth that suits all of them. Intensity features are comparable neither across

exposures nor, as shown above, across noise levels, and the degradation models are synthetic and independent where a real microscope delivers blur, shot noise and contrast loss together.

Three extensions follow: a per-image adaptive h scaled by the image’s own estimated median nucleus radius, which addresses the worst of those limitations; a second dataset with controlled ground truth (BBBC005 synthetic cells, considered and left out of scope), to separate what belongs to this method from what belongs to this collection; and the same evaluation harness applied to a learned segmenter, giving the classical-versus-learned comparison on identical splits and metric definitions that Section 3.2 explicitly cannot.

Citation

Caicedo, J.C., Goodman, A., Karhohs, K.W., Cimini, B.A., Ackerman, J., Haghighi, M., Heng, C., Becker, T., Doan, M., McQuin, C., Rohban, M., Singh, S., and Carpenter, A.E. (2019). Nucleus segmentation across imaging experiments: the 2018 Data Science Bowl. *Nature Methods* **16**(12), 1247–1253. doi:10.1038/s41592-019-0612-7

Ljosa, V., Sokolnicki, K.L., and Carpenter, A.E. (2012). Annotated high-throughput microscopy image sets for validation. *Nature Methods* **9**(7), 637 — the Broad Bioimage Benchmark Collection, which distributes this data as BBBC038v1: bbbc.broadinstitute.org/BBBC038